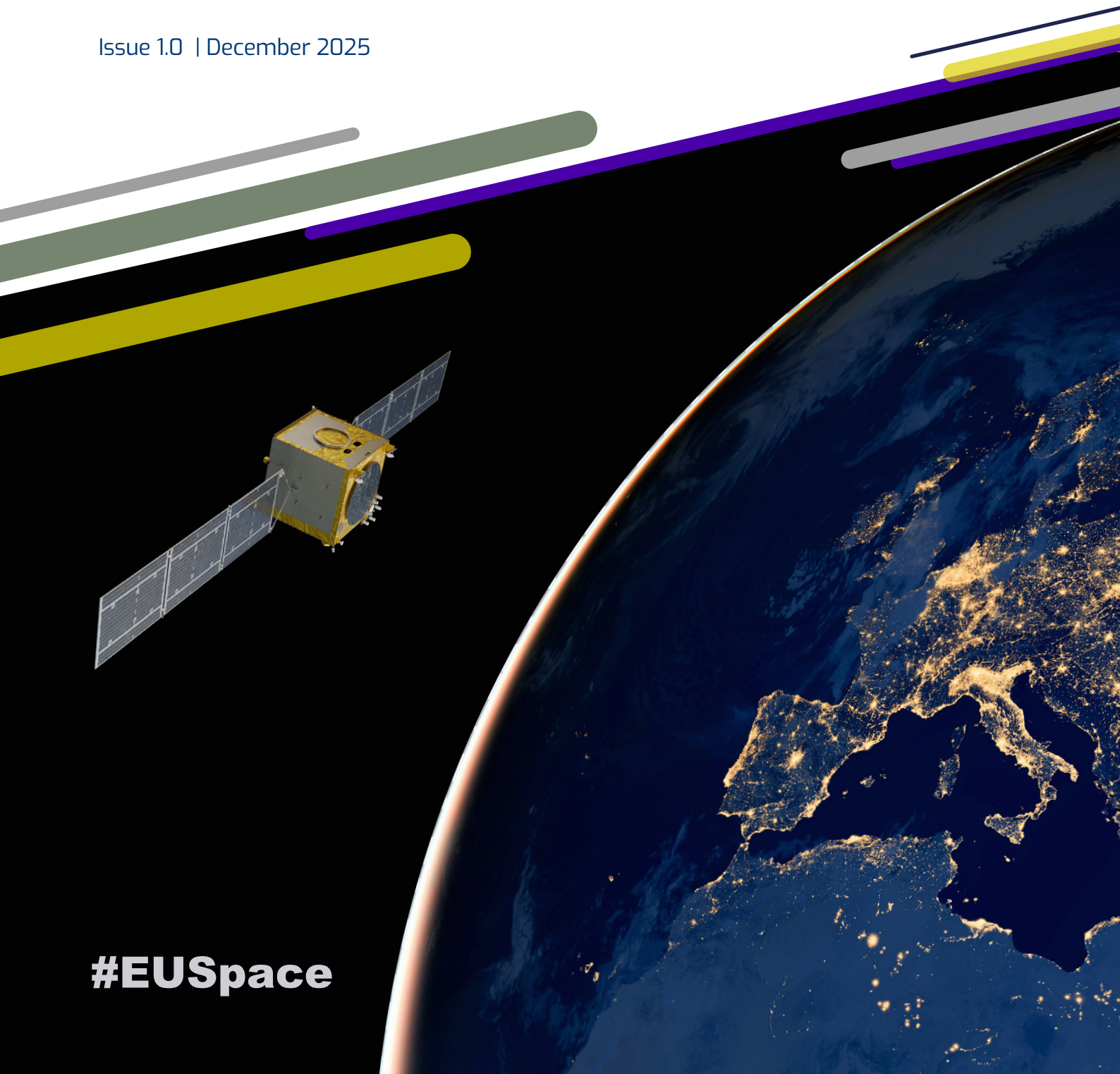
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GALILEO SECOND GENERATION (G2) EVOLUTION OF QUASI-PILOT SIGNALS AND INTERFACE CONTROL INFORMATION

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1 INTRODUCTION

Galileo Quasi-Pilot (QP) signals are designed to enable low complexity acquisition of the Galileo signals, to provide the capability to resolve quickly an initial receiver time ambiguity (up to 2.4 sec) using only the information carried by a QP signal as well as to support robust tracking of other Galileo signal components. The QP signals as described in this document are foreseen to be transmitted by Galileo 2nd Generation (G2G) satellites. It is reminded here that a first implementation of the QP (as identified in OS SIS ICD Issue 2.2 [RD.1]) is going to be provided already through a subset of Galileo FOC satellites, . The QP signal broadcast by the Galileo FOC satellites, labelled as E5a-QP and to be transmitted in the E5 band will benefit an expected 1 or 2 generations of mass-market chipsets (that have a typical lifetime of 2 to 5 years) before the G2G QP signal components will start being available. It should be noted that the E5a-QP does not provide all features of G2G QP. The objective of the QP is to support mass market (i.e. smartphones, IoT, drones and automotive) chipsets, as well as other markets in achieving faster Time-To-First-Fix (TTFF). Moreover, other professional markets can also benefit from this new feature.

This document is published for the benefit of Galileo users and receiver manufacturers, providing information on the planned G2G QP signals before their official release through a future update of the Galileo OS SIS ICD, and enabling receiver manufacturers to plan development with a clear understanding of the deployment of QP signals so as to ensure a smooth transition during their introduction. This information is released as a complement of the Issue 2.2 of the OS SIS ICD [RD.1] in which the technical specification of the new E5a-QP signal is published, in order to provide a complete information on the Galileo programme plans regarding the Quasi-Pilot signals.

The E5a-QP, described in the Galileo OS SIS ICD (starting with issue 2.2), will not be transmitted by G2G satellites, being superseded by the new QP signals. This means that the functionalities and availability of QP signals will progressively evolve with the launch of G2G satellites and gradual replacement of Galileo FOC satellites.

The main features provided by the Quasi-Pilot (QP) signal components are:

- low-complexity acquisition, enabling the fast access to the Galileo signals, including both legacy and new signals introduced through the G2G satellites.
- fast time ambiguity resolution through a so-called Time Dissemination Scheme (TDS).
- robust tracking of other new signals to be introduced through the G2G satellites, including new data signal components[1].

Those features are meant to support fast Time To First Fix (TTFF) for OS and other Galileo services supported by the OS signals. During the progressive deployment of G2G satellites, users will already benefit from faster time ambiguity resolution provided by the G2 QP signals, and thus faster TTFF, by having

[1] G2G data components will be provided in the different Galileo bands and will be transmitted at the same carrier frequency with a corresponding QP signal component. The G2G data components are meant to transmit a flexible data message with high data rate, including to enable enhanced Time to First Fix Data (TTFFD) and data demodulation robustness. The data components are complementing the different services with specific added-value data. An overview on the design of those new data components is provided in [RD.2]

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just one single G2G satellite within the first satellites being acquired. This document provides detailed information for the following QP signals:

- E1 Quasi Pilot 1 (labeled as E1-QP1), to be transmitted at 1576.443 MHz
- E1 Quasi Pilot 2 (labeled as E1-QP2), to be transmitted at 1597.926 MHz
- E5 Quasi Pilot (labeled as E5c-QP), to be transmitted at 1182.588 MHz

Future updates of Galileo SIS ICDs will also include a specific QP realization for the Galileo E6 signal baseline in support to Galileo HAS and SAS services.

This document provides an early information on the requirements and interface specifications necessary to implement the signal at receiver level and follows the structure of the Galileo OS SIS ICD, to easily map the elements provided here for the QP signal components with the legacy Galileo signal baseline elements provided in the ICD.

Although the elements presented in this document are at an advanced stage of development, the Galileo programme, as the Publishing Authority, makes no commitment that these elements will remain unchanged prior to their official publication in a future update of the OS SIS ICD.

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2 QUASI-PILOT SIGNAL COMPONENTS CHARACTERISTICS

Within this chapter the main signal parameters for the QP signal components are provided.

2.1 E1 Signal baseline

As previously mentioned, two QP signal components at different carrier frequencies have been identified for the E1 band, E1-QP1 and E1-QP2.

The benefits and use cases of E1-QP1 are currently under assessment and the provision of this signal is pending final confirmation. The modulation parameters for the E1 Quasi-Pilot signal components (E1-QP1 and E1-QP2, respectively) are specified in the following table.

Component	Carrier Frequency (MHz)	Modulation Type	Ranging Code Chip-Rate (MChip/s)	Code Length (chips)
E1-QP1	1576.443	**BPSK**	1.023	1023
E1-QP2	1597.926	**BPSK**	1.023	1023

Table 1: Baseline E1-QP Signal Parameters

It is to be noted here that an alternative modulation is being considered for the E1-QP2 signal, as specified in the next table.

Signal Component	Carrier Frequency (MHz)	Modulation Type	Ranging Code Chip-Rate (MChip/s)	Sub-carrier Frequency (MHz)	Code Length (chips)
E1-QP2	1597.926	BOC	1.023	1.023	1023

Table 2: Alternative E1-QP2 Signal Parameters

An assessment based on various use cases is being performed to decide whether it is worth considering changing the E1-QP2 modulation from BPSK(1) to BOC(1,1). The final specification of the E1-QP2 parameters will be described in the Galileo OS SIS ICD.

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2.2 E5 Signal baseline

The modulation parameters for the E5c-QP signal are specified in the following table.

Signal Component	Carrier Frequency (MHz)	Modulation Type	Ranging Code Chip-Rate (MChip/s)	Code Length (chips)
E5c-QP	1182.588	**BPSK**	1.023	1023

Table 3: E5c-QP Signal Parameters

2.3 Received Power Levels on Ground

The next table provides the expected minimum and maximum power levels on ground for QP signal components. The assumptions behind both minimum and maximum power levels provided in OS SIS ICD [RD.1] apply also here.

Signal Component	Total Received Minimum Power (dBW)	Total Received Maximum Power (dBW)
E1-QP1	-159.00	-153.75
E1-QP2	-159.00	-153.75
E5c-QP	-158.25	-153.0

Table 4: QP signal components Power on Ground

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3 SPREADING CODES CHARACTERISTICS

Within this chapter the characteristics of the QP signal components spreading codes characteristics are presented.

The primary codes for E1-QP1, E1-QP2 and E5c-QP are Gold Codes of length 1023 chips. Gold codes are a specific type of pseudo-random binary sequence generated by combining two linear feedback shift register (LSFR) sequences to produce well-defined correlation properties. Please refer to the OS SIS ICD for the generation of PRN sequences based on linear shift registers.

3.1 Parameters for the generation of QP Primary codes with linear feedback shift registers

QP primary codes are Gold Codes based on linear shift registers with the configuration given in the following table.

Length of shift registers (polynomial order)	Feedback tap polynomials (Octal)	
	Feedback 1	**Feedback 2**
10	2415 – Octal (Feedback taps: [2,3,8,10])	2443 – Octal (Feedback taps: [1,5,8,10])

Table 5: QP primary code LFSR parameters

3.2 Base register 1 start value for QP Primary PRN Codes

The start values for all base register 1 cells, in logic level notation, is '1' for all QP codes.

3.3 Base register 2 start value for QP Primary PRN Codes

The start values for all base register 2 cells, in logic level notation, is '1' for all QP codes.

3.4 Shift of register 2 for QP Primary PRN Codes

The output of register 2 is shifted by the shift index identified in the following table before XOR with the output of register 1. The following table provides the shift index for 50 QP PRN codes – additional PRN codes may be defined by the Galileo Program for QP use.

Code No	QP Shift Register 2	Code No	QP Shift Register 2
1	344	**26**	7

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Code No	QP Shift Register 2	Code No	QP Shift Register 2
2	337	**27**	270
3	268	**28**	259
4	397	**29**	929
5	879	**30**	921
6	942	**31**	200
7	956	**32**	407
8	51	**33**	897
9	988	**34**	885
10	31	**35**	934
11	255	**36**	680
12	279	**37**	531
13	444	**38**	774
14	992	**39**	210
15	147	**40**	162
16	896	**41**	817
17	663	**42**	711
18	33	**43**	925
19	592	**44**	477
20	19	**45**	201
21	96	**46**	170
22	237	**47**	225
23	626	**48**	23
24	475	**49**	924
25	123	**50**	117

Table 6: Base register 2 output shift

The assignment of specific codes to satellites will be detailed in future OS SIS ICD updates.

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4 MESSAGE STRUCTURE

This chapter provides details on the QP signal components message structure, describing in particular the Time Dissemination Scheme (TDS) used to enable rapid synchronisation to the Galileo System Time (GST), as well as the Quasi Pilot Health Status (QHS). The QHS is introduced to provide relevant signal health status information to users that might just use the QP signal component without continuously demodulating the Galileo navigation message.

The E1-QP1, E1-QP2 and E5c-QP primary codes are modulated by different overlay codes, depending on the information to be carried (TDS or QHS). As defined in the table below, in the case the signal is healthy, a TDS overlay code is used and further encoded by Variable Transition Phasing (VTP) symbols (as explained later). Otherwise, QHS overlay codes are used to flag the specific health status.

Overlay Code	Quasi-Pilot Health Status (QHS)	VTP symbols
TDS	Signal OK	Yes
QHS1	Signal out of service	No
QHS2	Spare	No
QHS3	Signal Component currently in Test	No

Table 7: Overlay codes for QP

The beginning of the overlay code is aligned with the beginning of the week.

4.1 Time Dissemination Scheme (TDS)

TDS is a specific feature of the QP signal components meant to allow the receiver, through a single QP signal, to rapidly synchronise to the GST enabling a faster TTFF. The design of the QP TDS has been driven by the target use cases, while trading-off the need to support the legacy Galileo signals with the possibility to enable stand-alone tracking of the new QP signal components. The TDS overlay code corresponds to a constant overlay sequence (i.e., all “1”). An overview of different options considered for the QP TDS and details on the design criteria are provided in [RD.3].

When the QP signal is healthy (“Signal OK” configuration), the resulting code sequence is modulated by Variable Transition Phasing (VTP) symbols, targeting the provision of a stand-alone time dissemination scheme providing fast time ambiguity resolution.

Each VTP symbol is formed by the sequence of two sub-symbols. The duration of the VTP symbol is fixed and equal to 240ms. However, the location of the transition between the 2 sub-symbols composing the VTP symbol is variable, and it is therefore used to encode the information. The location of the transition between sub-symbols is actually encoding the information, and can be {40, 44, … , 116} ms from the start of the VTP symbol. The following figure illustrates the concept.

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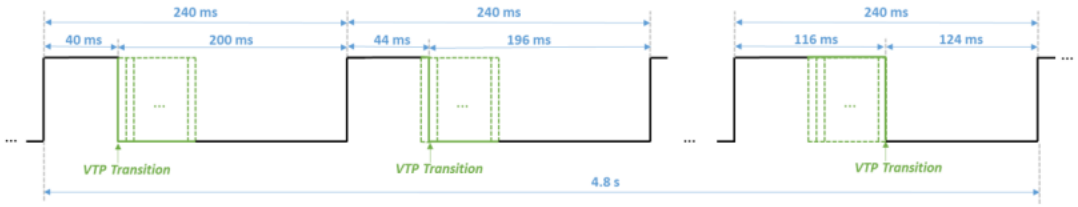


Figure 1: Illustration of TDS

The distance between two consecutive symbol transitions (corresponding to the length of one of the two sub-symbols of the VTP symbol) allows to identify the state transmitted by the VTP symbol. This is possible thanks to the asymmetry of the lengths of the two VTP sub-symbols for each of the 20 possible transmitted states.

With the QP TDS a time ambiguity of ±2.4 s can be resolved in less than 248 ms.

VTP symbol State [dec] transmitted by the overlay code	4.8-s time [s]	VTP sub-symbols length [ms]
#0	0.24	{40, 200}
#1	0.48	{44, 196}
#2	0.72	{48, 192}
#3	0.96	{52, 188}
#4	1.2	{56, 184}
#5	1.44	{60, 180}
#6	1.68	{64, 176}
#7	1.92	{68, 172}
#8	2.16	{72, 168}
#9	2.4	{76, 164}
#10	2.64	{80, 160}
#11	2.88	{84, 156}
#12	3.12	{88, 152}
#13	3.36	{92, 148}
#14	3.6	{96, 144}
#15	3.84	{100, 140}
#16	4.08	{104, 136}
#17	4.32	{108, 132}

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VTP symbol State [dec] transmitted by the overlay code	4.8-s time [s]	VTP sub-symbols length [ms]
#18	4.56	{112, 128}
#19	4.8	{116, 124}

Table 8: VTP symbol states and correspondance to 4.8 sec time intervals

4.2 Quasi-Pilot Health Status (QHS) Overlay Codes

For the transmission of the QP health status, the transmission of alternative overlay symbol rates is used to indicate a health status different from ”signal OK” (in this case no VTP symbols are transmitted). The overlay code sequences for the Quasi-Pilot Health Status (QHS) in E1-QP1, E1-QP2 and E5c-QP are defined in the following table. The overlay codes QHS1, QHS2, and QHS3 in the following table represent the code sequence {1,0}.

Code identifier	Code length	Number of hexadecimal symbols	Code sequence (Hexadecimal)
QHS1	64	16	FFFFFFFF00000000
QHS2	48	12	FFFFFF000000
QHS3	40	10	FFFFF00000

Table 9: QHS overlay code sequences for QP

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5 LIST OF REFERENCES

RD.1 **The European GNSS (Galileo) Open Service Signal-In-Space Interface Control Document** (OS SIS ICD), Issue 2.2, November 2025, https://www.gsc-europa.eu/electronic-library/programme-reference-documents

RD.2 J.A. Garcia-Molina, S. Wallner, J. Miguez, J. Parro, F.T. Melman, C. Vazquez Alocen, G. Da Broi, G. Lopez-Risueno, V. Lucas-Sabola, I.P. Peppa, O. Cervelló i Nogués, F. Formaggio, F. Soualle, J.J. Floch, D. Peron, F. Pattaro, M. Ceccato, F. Moretto, **Galileo Data Signals: Enabling Fast-TTFF and Flexibility with the GeoFocus and GNSE Schemes**, ION GNSS+ 2023

RD.3 J.A. Garcia-Molina, S. Wallner, F.T. Melman, C. Vasquez Alocen, G. Da Broi, J.M. Parro, J. Hahn, G. Lopez-Risueno, G. Houot, J.A. del Peral-Rosado, V. Lucas, O. Cervelló i Nogués, F. Formaggio, M. Vergara, A. Waelkens, F. Soualle, J.J. Floch, M. Dotzauer, M. Ouedraogo, M. Paonni, M. Caparrini, **Galileo Quasi-Pilot Signals: Assessment and Design Options for Acquisition and Time Dissemination**, ION GNSS+ 2023

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